TFPT — Appendix H: the horizon unit system

One seam constant $c_3 = \frac{1}{8\pi}$ as the universal horizon thermal code
(Hawking, de Sitter, Unruh, Page, scrambling — *standard physics in seam units, not new gravity*)

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What this document is about

A change of bookkeeping, not new gravitational physics: if gravity is the geometry-channel readout of the seam, then *all* horizons read the same boundary constant $c_3 = \frac{1}{8\pi}$. This note collects the readouts — Hawking, de Sitter and Unruh temperature, black-hole thermodynamics, the Page time, scrambling, the Nariai bound, $v_{\mathrm{GW}} = c$ and cosmic birefringence — in seam units, with two genuine compiler fingerprints ($1920 = |W(D_5)|$, $|\mu_4| = 4$). The Hawking normalisation itself is since *measured* on the seam: the reconstructed free OS quotient is KMS with $\beta_{\mathrm{angle}} = 2\pi$ exact, $T_{\mathrm{seam}} = 4c_3$ ([`verification/v526_seam_thermal_kms_nariai_bridge.py`]) — temperature joins geometry and anomaly as the *third leg* of c_3 ([C]-typed reading; the entropy-fraction bridge honestly does not close).

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The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\mathrm{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions** (plus one standalone working note), best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to,

and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

Working notes: N1. `note_e8_gaussian_code` — the Gaussian code bridge: E_8 over $\mathbb{Z}[i]$ via Construction A over the extended Hamming code $[8,4,4]$, the canonical four-bit quotient $L/(1+i)L \cong \mathbb{F}_2^4$, and the G_{31} quartic companion (exact algebra, machine-certified).

You are reading companion H (Appendix H — Horizon Readouts).

TFPT status at a glance — read this card the same way in every document

Two structural interfaces, one compiler. P1 is a primitive reflection-positive seam kernel with unit winding; its normalization is $c_3 = \frac{1}{8\pi}$. From this datum and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2), the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The numerical readouts c_3 and g_{car} are elementary symmetric readouts of the anchor $a = (1,1,2)$. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below. Existing reductions share the four-mark geometry and anchor $a = (1,1,2)$; both W-bridge premises are derived at finite level (`v1014/v1015`), modulo the external MMST identification. The additional P1 determinant-response identification is separate. The Architecture chapter "Shared origin" consolidates these scopes (`REC20261105C3`).

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); CLOSED METROLOGY ($\mathbb{R}_+$ torsor; one dim. calibration by dimensional analysis; No-Unit v153/v725)	closed metrology [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Beside that compiler rest sits the strict-physical-TOE rest: the ten-component Rest_{TOE} / the eight gates of `TFPT.TOE.COMPLETE.01` [O] ($\text{TOE}_{\text{Complete}} :=$ AND(T1..T8); not a closure).

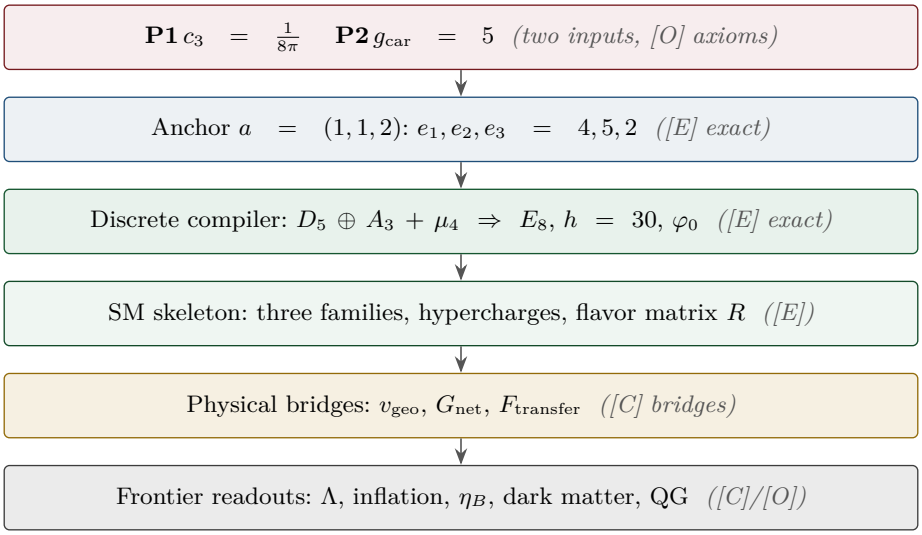
The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]. The classical metric-sector field equation is no longer only an $R{+}R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2|2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit v_{geo}; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. The lattice and index witnesses and the cited scaling and OS theorems establish the stated conditional route; they do not

leave every old prerequisite untouched. The later charged MMST manuscript proves its uniform Green, edge graph, crossed-product and relaxed TEL-B estimates and retains the named microscopic local-algebra exhaustion condition for the full source. Its P1 corollary separately requires identification of the determinant-line response with the additive transitive four-channel response. The declared operational seed already supplies state and time data. The physical question concerns applicability of these theorems to the same source. The unconditional seam parent and T1–T8 keep their ledger statuses. Under the conformal-net hypotheses, rotation invariance makes the geometric deck a consequence (`v335`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76/v330`). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels $(U_{\mathrm{wall}}, G_{\mathrm{metric}}, F_{\mathrm{frontier}})$ are retained only for ledger continuity — the live residual is $v_{\mathrm{geo}} \oplus F_{\mathrm{transfer}}$, with the seam G_{net} closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test



Scope and honest typing. If gravity is the geometry-channel readout of the seam (companion note "Seam response grammar"), then horizons are not separate miracles — they all read the *same* boundary normaliser $c_3 = 1/(8\pi)$. This note collects the readouts. Markers: [C] is standard horizon physics rewritten in TFPT/seam units (no new claim) or a search target; [E] is an exact compiler connection (audit-level) or a reproduced numerical readout; [O] is a search target, not a result. Nothing here is sold as new gravitational physics; it is a *change of bookkeeping* that exposes shared structure.

Unit conventions used throughout

Three unit systems appear; the seam factor $c_3 = 1/(8\pi)$ is the same number in all of them, only the carried dimensionful constants change:

system	convention	Hawking temperature
Planck	$G{=}\hbar{=}c{=}k_B{=}1$	$T_H = c_3/M = 1/(8\pi M)$
reduced Planck	$\bar{M}_{\rm Pl} = M_{\rm Pl}/\sqrt{8\pi}$, $\hbar{=}c{=}k_B{=}1$	$T_H = c_3\,\bar{M}_{\rm Pl}^2/M$
SI	restore $G, \hbar, c, k_B$	$T_H = \dfrac{\hbar c^3}{8\pi G k_B M} = c_3\,\dfrac{\hbar c^3}{G k_B M}$

All boxed formulas below are in Planck units unless stated; the seam constant is dimensionless.

Standard references for the formulas reframed here: Hawking temperature and radiation (Hawking 1975); Bekenstein–Hawking entropy (Bekenstein 1973, Hawking 1975); Page time (Page 1993); fast scrambling $t_{\rm scr} \sim \beta \log S$ (Sekino–Susskind 2008, Maldacena–Shenker–Stanford 2016); Nariai bound (Nariai 1950); de Sitter thermodynamics (Gibbons–Hawking 1977). *Nothing below is a new derivation of these* — only a rewriting in the seam unit c_3.

1 The seam constant is the universal horizon temperature factor

Every Killing horizon has $T = \hbar\kappa/(2\pi c k_B)$. Since

$$\boxed{\frac{1}{2\pi} = 4c_3, \qquad \frac{1}{8\pi} = c_3} \qquad \textbf{[E]},$$

the universal factor *is* the seam constant: $T_{\rm hor} = 4c_3\,\hbar\kappa/(ck_B)$. Black holes, de Sitter and Unruh therefore share *one* thermal grammar. **[C]** [verification/v8_horizon.py]

2 Schwarzschild thermodynamics in four c_3-lines

For a non-rotating, uncharged hole in Planck units ($G{=}\hbar{=}c{=}k_B{=}1$):

$$\boxed{T_H = \frac{c_3}{M}, \qquad S_{BH} = \frac{M^2}{2c_3}, \qquad P_H = \frac{c_3}{1920\,M^2}, \qquad \tau_{\rm evap} = \frac{640}{c_3}M^3} \qquad \textbf{[C]}.$$

Two of the constants are compiler numbers, not coincidences:

- the Hawking temperature factor is $c_3 = 1/(8\pi)$ — the same P1 boundary normaliser; **[E]**
- the radiated-power denominator is $1920 = |W(D_5)|$, the Weyl-group order of the D_5 carrier. **[E]**

Scope of P_H: this is the idealised massless single-channel blackbody normalisation. Greybody factors and the radiated species content shift the phenomenological evaporation coefficient; the c_3-form is the convention statement, not a general phenomenological prediction.

object	T_H	lifetime / entropy
solar-mass hole	6.17×10^{-8} K	$\tau_{\rm evap} \approx 2.1 \times 10^{67}$ yr; $S/k_B \approx 1.05 \times 10^{77}$
PBH evaporating now (1.7×10^{11} kg)	$k_B T \approx 61$ MeV	relevant for γ/ν PBH signatures

The temperature is modular; the entropy carries the scalaron. [verification/v208_bh_thermodynamics.py] The Hawking factor is not put in by hand: the stationary exterior modular flow $\Delta^{it} = e^{-2\pi t K_H}$ (Bisognano–Wichmann / Tomita–Takesaki) makes the outside state KMS at inverse temperature 2π, so $T_H = \kappa/(2\pi)$ — and that 2π *is* the seam unit $1/(4c_3)$, reproducing $T_H = c_3/M$

above (the seam=horizon modular identification ties to $[\rho, \Lambda_\Sigma] = 0$, [verification/v198_modular_commutator_reduction.py]). The same normalisation is now *measured* on the reconstructed free seam OS quotient (`SEAM.THERMAL.KMS.01`, [verification/v526_seam_thermal_kms_nariai_bridge.py]): detailed balance gives $\beta = N$ clock steps at both levels, hence $\beta_{\text{angle}} = 2\pi$ and $T_{\text{seam}} = 4c_3$ — temperature joins geometry and anomaly as the *third leg* of c_3 ([C]-typed reading: seam euclidean circle = thermal circle of the reconstructed horizon dynamics; the entropy-fraction bridge to the Nariai ledger honestly does not close). The induced $R + R^2$ scalaron then corrects the area law to the Wald entropy

$$S_W = \frac{f_R\, A}{4G} = \frac{A}{4G}\Big(1 + \frac{R_h}{3M_s^2}\Big), \qquad M_s = c_3^{7/2}\bar{M}_{\text{Pl}},$$

an *exact* consequence of $f(R) = R + R^2/(6M_s^2)$ ([verification/v28_gravity_fR.py]); the leading $A/4G$ is the c_3 area law ($1/4 = 1/|\mu_4|$). [E] for the identities; the black-hole/modular identification is [C].

3 Page time and scrambling

Since $S_{BH} \propto M^2$ and $\tau \propto M^3$, the Page point ($M \to M/\sqrt{2}$) is

$$t_{\text{Page}} = \Big(1 - \tfrac{1}{2\sqrt{2}}\Big)\tau_{\text{evap}} \approx 0.6464\,\tau_{\text{evap}} \qquad \text{[C]}.$$

In TFPT the Page curve is *boundary recoverability*: the physical sector is OS-reconstructed and gapped, and the recovered mutual information decays at the transfer rate, $I(A{:}C|B)_n \sim \lambda_2^n = (2/3)^{6n}$. **This is the same operator that builds the Standard Model:** the boundary transport spectrum is $\{1, (2/3)^6, (1/3)^6\}$ and its sub-leading eigenvalue $\lambda_2 = (2/3)^6$ is exactly the flavor-gap eigenvalue ($\Delta_{\text{gap}} = -\log(2/3)^6 = 6\log\frac{3}{2}$). One boundary transport thus governs *both* the SM flavor hierarchy and the horizon information recovery. [E] [verification/v54_seam_horizon_keystones.py]

The recovery is a finite code ([verification/v221_seam_qecc.py]). Read as a quantum channel, this is a finite recoverability (error-correction-style) code: the code space is the carrier half-spinor $\dim S^+ = 16$, the gapped transport T on the cusp-weight 3-space (deviation directions $(1,-1,0)$ and the Nariai anchor $(1,1,-2)$) is symmetric and doubly stochastic (CPTP), and on the trace-zero subspace it contracts with operator norm exactly $(2/3)^6$, so $\|T^n\delta\| \le (2/3)^{6n}\|\delta\|$ — a Knill–Laflamme-type per-step recovery bound, with a free ratio breaking it. "Finite recoverability code" is [E]; the explicit Petz recovery map and the rank-one (one-dimensional baby-universe, cf. Engelhardt 2025) contraction limit $\|T^n - P_\infty\| = (2/3)^{6n}$ are now realised numerically (`experiments/recovery-channel`), so the Petz identification of T with the Page recovery map is verified at the channel level [C]; the full holographic reading stays gated on the Seam–Horizon theorem. The scrambling prefactor carries a μ_4 signature: with $\beta = 1/T_H = 8\pi M$, $\beta/2\pi = 4M$, so

$$\boxed{t_{\text{scr}} \sim |\mu_4|\, M \log S} \qquad (|\mu_4| = 4) \qquad \text{[E]-audit.}$$

4 De Sitter and the Nariai bound from the Λ closure

With the action-grammar Hubble/Λ readouts ($H_0/\bar{M}_{\text{Pl}} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$, $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 \sim e^{-2\alpha_\star^{-1}}$), the asymptotic de Sitter horizon gives

$$\boxed{\frac{T_{dS}}{\bar{M}_{\text{Pl}}} = 16c_3^2\, e^{-\alpha_\star^{-1}}, \qquad S_{dS} = \frac{e^{2\alpha_\star^{-1}}}{128c_3^4} = 32\pi^4 e^{2\alpha_\star^{-1}} \approx 3.32 \times 10^{122}} \qquad \text{[C]}.$$

This is the area-law horizon capacity of the universe from $(\alpha_\star^{-1}, c_3, H_\Lambda)$ alone. The maximal black hole in de Sitter (Nariai) is

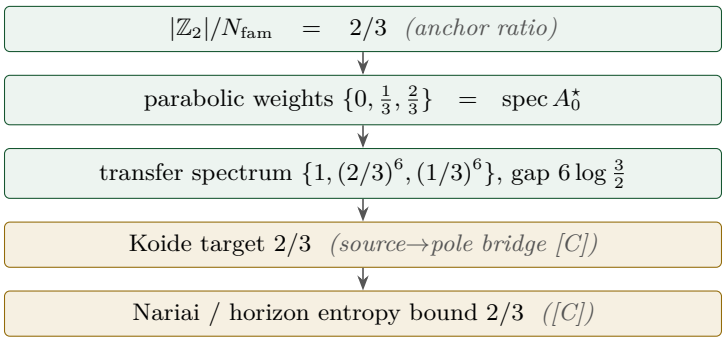
$$M_N \approx 2.16 \times 10^{22}\, M_\odot \approx 4.3 \times 10^{52}\,\text{kg}, \qquad S_N = \tfrac{1}{3}S_{dS} \qquad \text{[C]},$$

the absolute gravitational-capacity limit, not an observed structure.

5 The maximal black hole is the anchor: SdS in seam units

The Nariai limit is not just a capacity number — written in compiler units, the entire Schwarzschild–de Sitter (SdS) family lands on atoms that are already load-bearing elsewhere in TFPT, with no adjustable parameter on either side. **[E]** [verification/v101_horizon_anchor.py]

The defect reading: no singularity, a seam fixed point ([verification/v209_bh_regular_core.py]). In compiler units the interior is *not* a curvature blow-up but the seam attractor: the same gapped transport whose sub-leading eigenvalue is $\lambda_2 = (2/3)^6$ ([verification/v56_unique_attractor.py]) drives $\varphi \to \varphi_\star$ ($\mathrm{d}\varphi/\mathrm{d}t = 0$), so "$\rho \to \infty$" is replaced by a fixed point; information returns through the same Page-recovery channel (λ_2 per step, [verification/v54_seam_horizon_keystones.py]); and the end state is a holographic Planck-scale floor ($S_{BH} = A/4$, one cell per $|\mu_4| = 4$ Planck areas), not a point. *Honest:* a **[C]** structural reading — the old RN/torsion-charge metric is *not* resurrected, it is superseded by the Nariai/seam=horizon anchor here.



Green: exact motif ([E]). Gold: physical bridge ([C]) — the same ratio, two epistemic types.

Nariai = the anchor; the Koide 2/3 is the entropy bound [E] ([verification/v101_horizon_anchor.py])

With $f(r) = 1 - 2M/r - \Lambda r^2/3$, the horizon cubic at the maximal mass $M_N = 1/(3\sqrt{\Lambda}) = 1/(N_{\mathrm{fam}}\sqrt{\Lambda})$ is, in units $1/\sqrt{\Lambda}$,

$$\boxed{t^3 - 3t + 2 = (t-1)^2(t+2)} \qquad \text{roots } (1, 1, -2),$$

with coefficients $(-3, 2) = (-N_{\mathrm{fam}}, |\mathbb{Z}_2|)$ — and the root pattern is exactly the **traceless projection of the anchor**: $a - \frac{p_1(a)}{3}\mathbf{1} = -\frac{1}{3}(1, 1, -2)$ for $a = (1, 1, 2)$ (compare $\chi_a = (t-1)^2(t-2)$, the anchor characteristic polynomial). Five exact companions:

- **Entropy bound = the Koide branch value.** Each Nariai horizon carries $S_{dS}/3 = S_{dS}/N_{\mathrm{fam}}$ and the total is $\boxed{S_{\mathrm{Nariai}}/S_{dS} = 2/3 = |\mathbb{Z}_2|/N_{\mathrm{fam}}}$ (deficit exactly $S_{dS}/3$). The interpolation in $x = r_b/r_c$ is the closed form $S_{\mathrm{tot}}/S_{dS} = (x^2{+}1)/(x^2{+}x{+}1)$ — denominator $= \Phi_3(x)$, the N_{fam} cyclotomic — monotone from 1 to 2/3 with its minimum exactly at the merge (Fig. 2a). The floor admits a *variation-free* one-line proof: $3(x^2{+}1) - 2\Phi_3(x) = (x{-}1)^2 \geq 0$, so $S_{\mathrm{tot}} \geq \frac{2}{3}S_{dS}$ with equality *iff* $x = 1$ (the Nariai merge) **[E]** ([verification/v190_nariai_entropy_bound.py]).
- **Three-sheet conservation.** The cubic is traceless with fixed e_2, so $\pi \sum_i r_i^2 = 6\pi/\Lambda = |\mathbb{Z}_2|\, S_{dS}$ for *every* M: the black hole redistributes entropy between the two physical sheets and the virtual third root; the three-sheet total never moves.
- **Koide-form quotient of the roots.** $Q_{\mathrm{geom}} = \sum r_i^2/(\sum |r_i|)^2$ runs monotonically over $[\frac{3}{8}, \frac{1}{2}]$: pure dS gives $\frac{1}{2} = |\mathbb{Z}_2|/|\mu_4| = \delta$ (the half-step), Nariai gives $\frac{3}{8} = p_2(a)/e_1(a)^2$ — exactly the two nonzero $SU(4)_1$ conformal weights (Fig. 2b).
- **The mass line is a double cover.** $\mathrm{disc}_r \propto (1 - 3m)(1 + 3m)$ in $m = M\sqrt{\Lambda}$: branch points $m = \pm 1/N_{\mathrm{fam}}$, separation $2/3 = |\mathbb{Z}_2|/N_{\mathrm{fam}}$, deck involution = the black-hole ↔ cosmological horizon swap — the gravitational twin of the flavor sheet $\mathbb{Z}_2$ and of the split N_{fam}-linear cover $(3x{+}2)(3x{+}5)$ of the flavor sector (Fig. 1). The two branch structures share one N_{fam}-linear ramification: the affine map $q = \frac{7}{2} + \frac{N_{\mathrm{fam}}^2}{2}m$ carries the SdS branch points $m = \pm 1/N_{\mathrm{fam}}$ onto the flavor branch labels $\{2, 5\} = \{|\mathbb{Z}_2|, g_{\mathrm{car}}\}$ (midpoint 7/2). This is an exact *relabeling* — the shared invariant is the 2/3 ramification, not the map, which exists for any two pairs of points — so it is recorded as

a [C] structural alignment, not a theorem ([verification/v191_universal_branch_line.py]).

- **Temperature lemma / orientation.** $|\kappa_b/\kappa_c| = (2x{+}1)/(x(x{+}2)) > 1$ for $x < 1$ (equality iff $x = 1$): the black-hole sheet is *always* hotter, so evaporation flows *away* from the merge — the anchor configuration is the **repeller** and the democratic endpoint the attractor, the same orientation as the flavor flow (Koide attractor / carrier repeller, the frontier note). *Audit reading.*

Completing the seam-unit table: $dM = c_3\,\kappa\,dA$ (first law), $M = 2c_3\,\kappa A$ (Smarr), $S \le ER/(4c_3)$ (Bekenstein), $P_H = c_3/(1920\,M^2)$, lifetime $\tau = 5120\pi M^3 = 128\,g_{\rm car}M^3/c_3$ ($128 = 2^7$ the dS prefactor, 7 the scalaron exponent; photon/geometric-optics conventions, typed external), Kerr $A_{\rm ext} = M^2/c_3$ with $A_{\rm ext}/A_{\rm Schw} = 1/2 = 1/|\mathbb{Z}_2|$ and $S_{\rm ext} = M^2/(4c_3)$, and the Bekenstein–Mukhanov–Hod area quantum $4\ln 3 = \ln 81 = \ln(N_{\rm fam}^4) = \ln(\text{disc of the flavor cover})$ ([C] as the QNM reading). *Honest scope:* the GR identities are textbook facts; the [E] content is six independent landings on already-load-bearing atoms with zero free parameters. The "carrier in the bulk" reading (two glue chiralities = two horizon sheets, ramification = Nariai) stays [C]. No new data test (M_N today $\sim 10^{22}M_\odot$): a structure surface, not a measurement surface.

The dual anchor: the inverse flavor response *is* the Nariai root [E] ([verification/v134_dual_anchor.py])

The Nariai pattern is not only the traceless anchor — it is stored *inside the flavor compiler* as a dual invariant. For the residue/winding operators R, L of the flavor sector and the anchor $a = (1,1,2)$,

$$\boxed{d := a^\top R^{-1} = a^\top L^{-1} = \left(-\tfrac{1}{2}, -\tfrac{1}{2}, 1\right)}, \qquad d\cdot\mathbf{1} = 0, \quad d\cdot a = 1, \quad (1,1,-2) = -2d,$$

i.e. $d = \frac{3}{2}\left(a - \frac{4}{3}\mathbf{1}\right)$ is the *normalised traceless anchor*, proportional to the Nariai double-root pattern above. The invariance is exact and structural (Sherman–Morrison): with $L = R + 6\,\mathbf{1}e_1^\top$, a covector is winding-invariant iff it annihilates $R^{-1}\mathbf{1} = \frac{1}{4}(1,1,-1)$ — the winding deformation lives in the democratic direction $\mathbf{1}$, which d kills. Controls: $\mathbf{1}$, e_1 and the torsion normal $n = (5,-9,6)$ do *not* lie on the invariance plane $(\frac{1}{4}, \frac{1}{4}, -\frac{5}{2})$, so the anchor's membership is special. Together (d, n) form the dual normal pair of the flavor boundary: d reads the traceless horizon structure, n reads first-generation torsion ($n^\top R = (8,0,0)$, $n^\top L = (20,0,0)$). *Typing:* all identities [E]; the reading — a third, purely *algebraic* leg of the flavor ↔ horizon bridge, beside the shared clock spectrum ([verification/v126_clock_wall_bridge.py]) and the entropy power law ([verification/v129_entropy_power_law.py]) — stays [C].

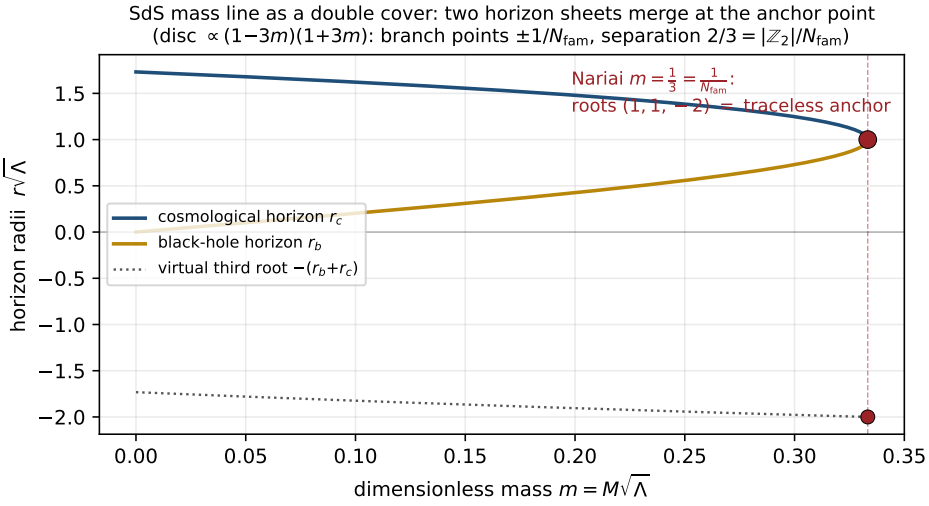


Figure 1: The SdS mass line as a double cover: black-hole and cosmological horizon sheets merge at the Nariai point $m = 1/N_{\rm fam}$, where the three roots form the traceless anchor pattern $(1,1,-2)$. [verification/v101_horizon_anchor.py]

One orientation: the anchor is the *stationary repeller* in both sectors ([E]/[C]). The evaporation direction of the temperature lemma upgrades to exact variational structure on both

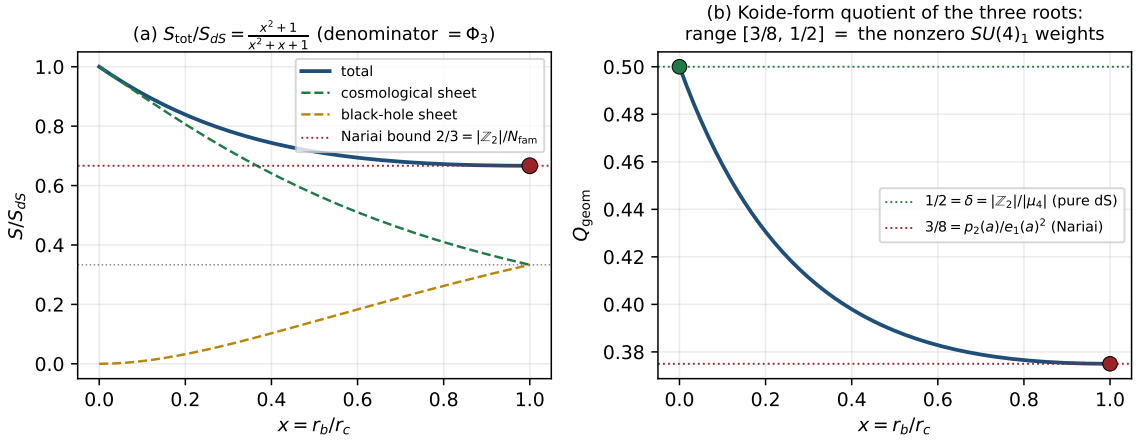


Figure 2: (a) Total SdS entropy interpolates from S_{dS} to the Nariai bound $\frac{2}{3}S_{dS}$ via $(x^2{+}1)/\Phi_3(x)$. (b) The Koide-form quotient of the three roots runs exactly over the nonzero $SU(4)_1$ weights $[\frac{3}{8}, \frac{1}{2}]$. [verification/v101_horizon_anchor.py]

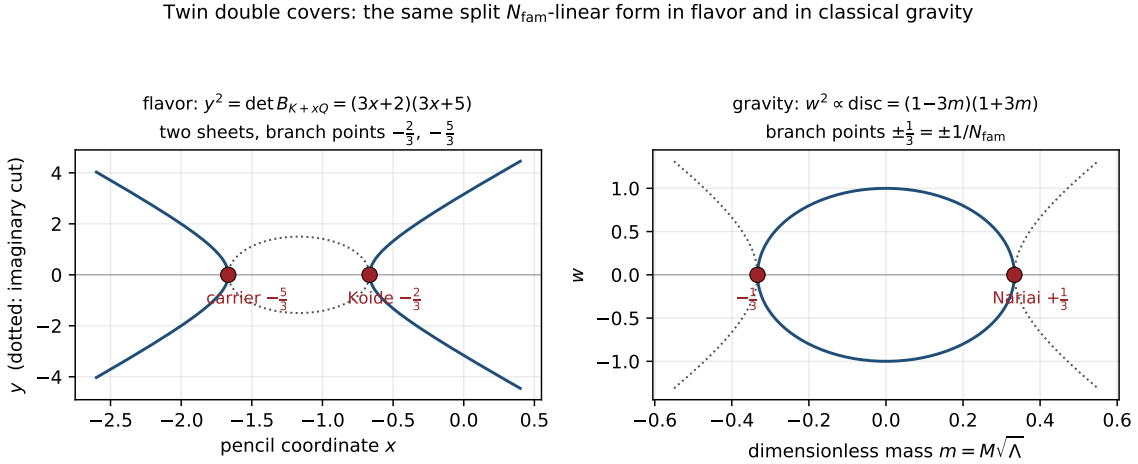


Figure 3: Twin double covers: the flavor anchor-block cover $y^2 = (3x{+}2)(3x{+}5)$ and the SdS mass-line cover $w^2 \propto (1{-}3m)(1{+}3m)$ share the split N_{fam}-linear form with rational branch points at spine-over-family values. [verification/v101_horizon_anchor.py]

one-dimensional moduli lines (Fig. 4). **Flavor:** the canonical relaxation $\dot{q} = (\Delta/N_{\text{fam}})(q{-}2)(q{-}5)$ is the *gradient flow* of the cubic potential

$$V(q) = -\frac{\Delta}{N_{\text{fam}}}\Big(\frac{q^3}{3} - \frac{7}{2}q^2 + 10q\Big), \qquad \boxed{V''(2) = +\Delta, \quad V''(5) = -\Delta}$$

— the critical points are exactly the two branch points, the stationary curvatures are ±the transfer gap, the inflection sits at $q = \frac{7}{2}$ = scalaron/2, and the Lyapunov rate is constant: $d(-\ln\rho)/dt = \Delta$ exactly. **Gravity:** the SdS entropy functional has

$$\frac{d}{dx}\frac{S_{\text{tot}}}{S_{dS}} = \frac{(x-1)(x+1)}{\Phi_3(x)^2}, \qquad \boxed{\Big(\frac{S_{\text{tot}}}{S_{dS}}\Big)''(1) = \frac{2}{9} = \frac{|\mathbb{Z}_2|}{N_{\text{fam}}^2} = \frac{2}{3}\cdot\frac{1}{3}}$$

— the Nariai/anchor point is the *unique* physical stationary point, its curvature is a grammar constant (= the product of the two Nariai entropy fractions), and the temperature lemma makes the evaporation an entropy *ascent* away from it. **Orientation theorem [E]:** in both sectors the anchor-type configuration is a stationary point of the natural functional and the dynamics

flows away from it, with grammar-constant stationary curvatures. *Honest disanalogies (recorded):* the flavor attractor is itself a branch point while the gravity attractor is the smooth endpoint; the flavor functional is the potential of a [C]-typed relaxation while the gravity functional is the Bekenstein–Hawking entropy — "one variational principle of the seam" stays a *reading* [C], and no open gate is touched. [verification/v102_seam_orientation.py]

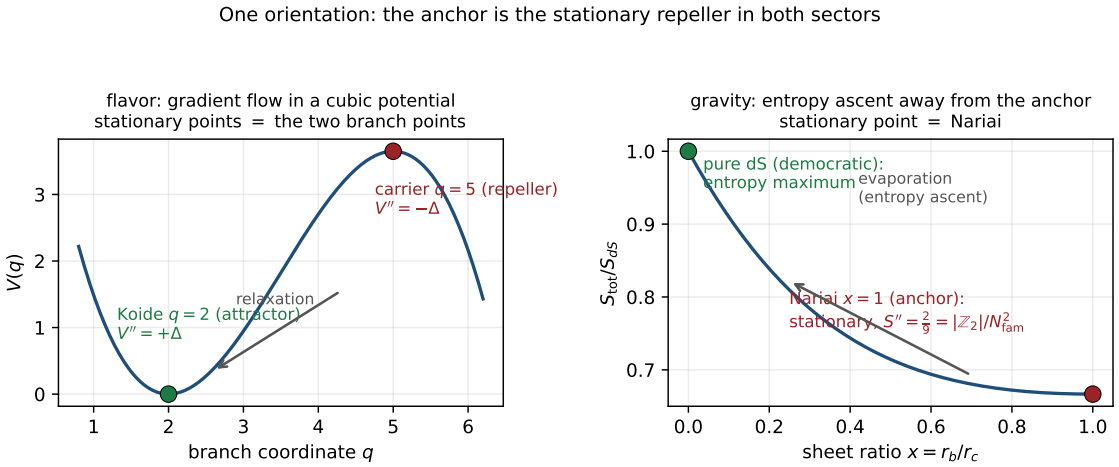


Figure 4: The orientation theorem: the anchor configuration is the stationary repeller of the natural functional in both sectors — flavor (cubic potential, curvatures $\pm\Delta$) and gravity (SdS entropy, curvature $2/9 = |\mathbb{Z}_2|/N_{\rm fam}^2$). [verification/v102_seam_orientation.py]

The trisection normal form: the canonical coordinate exists [E] ([verification/v103_trisection_normal_form.py])

The curvature-matching question of the previous box has a closed answer. The SdS horizon cubic is uniformized by **angle trisection**: with $r = 2\cos\theta$ (units $1/\sqrt{\Lambda}$), $r^3 - 3r = 2\cos 3\theta$ *exactly*, so the cubic is $\cos 3\theta = -3m$ — the three horizon roots are the three trisection branches (the $\mathbb{Z}_3$ rotation = the triality deck, cf. $\operatorname{coker} Q = \mathbb{Z}/N_{\rm fam}$). In the centered angle ψ ($\cos\varphi = -3m$, $\psi = \varphi - \pi$; Nariai $\psi = 0$, pure dS $\psi = \pi/2$, deck $\psi \to -\psi$):

$$m = \frac{\cos\psi}{N_{\rm fam}}, \qquad \boxed{\frac{S_{\rm tot}}{S_{dS}} = \frac{4}{3} - \frac{2}{3}\cos\frac{2\psi}{3}}$$

— *one* cosine whose mean ($|\mu_4|/N_{\rm fam}$), amplitude and frequency (both $|\mathbb{Z}_2|/N_{\rm fam}$) are glue atoms over the family count (Fig. 5). Three exact consequences:

- **Canonical curvature = the Koide constant to the family power:** $\sigma''(\psi{=}0) = \frac{8}{27} = (\frac{2}{3})^3 = (|\mathbb{Z}_2|/N_{\rm fam})^{N_{\rm fam}}$.
- **Invariant base slope:** $d\sigma/dm|_{\rm Nariai} = -\frac{8}{9} = -\operatorname{rank} E_8/N_{\rm fam}^2$ (coordinate-invariant, cross-checked in both coordinates; generally $-\frac{4}{3}\sin(2\psi/3)/\sin\psi$); dimensionful $dS_{\rm tot}/dM|_N = -r_N/(N_{\rm fam}c_3)$.
- **The bridge:** the flavor invariant is a *rate*, multiplier $(2/3)^6 = (2/3)^{2N_{\rm fam}}$ per transport step; the gravity invariant is a *curvature*, $(2/3)^{N_{\rm fam}}$ in the canonical angle — same base $2/3 = |\mathbb{Z}_2|/N_{\rm fam}$, exponent ratio $2 = |\mathbb{Z}_2|$. What is still missing for a full identification is the gravity-side *clock* (the near-Nariai evaporation generator that converts curvature into a rate) — the same class of missing object as the Koide transfer generator (frontier note): **one clock, two known geometries**. That identification stays [C].

The classical clock speaks anchor — and the honest $(2/3)$-test ([E]/[C]). The clock question of the previous box has a *classical* half, and it is pure GR (Ginsparg–Perry 1983; Bousso–Hawking class): linearize around the Nariai geometry $dS_2 \times S^2$. **Static pin [E]:** on the dS_2 static patch the function $\phi(\rho) = \rho$ satisfies $\frac{d}{d\rho}[(1-\Lambda\rho^2)\phi'] = -2\Lambda\phi$ *exactly* — and this static mode *is* the exact SdS

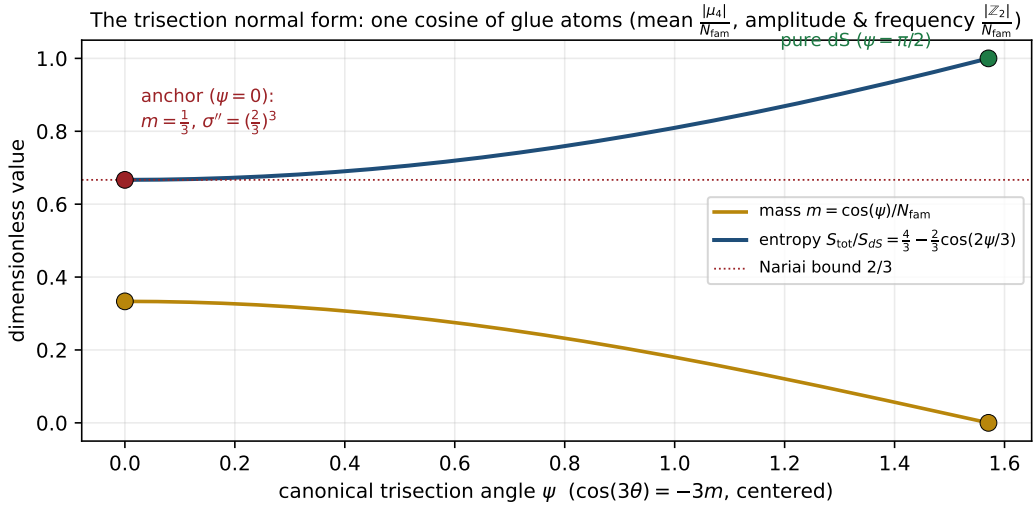


Figure 5: The trisection normal form: in the canonical angle ψ the mass and the total entropy are pure (co)sines built from glue atoms; the anchor sits at $\psi = 0$ with canonical curvature $(2/3)^3$. [verification/v103_trisection_normal_form.py]

two-horizon deformation, so the S^2-radius modulus mass is pinned internally: $m^2 = -2\Lambda = -|\mathbb{Z}_2|\Lambda$ (the Laplace-type form of the linearization is the one external GR import, typed honestly). The Euclidean spectrum is the Ginsparg–Perry tower $(l(l+1)-2)\Lambda$: exactly *one* negative mode $-|\mathbb{Z}_2|\Lambda$. **The clock [E]:** in Hubble units the homogeneous modes $\phi \sim e^{\lambda H t}$ obey

$$\boxed{\chi_{\rm clock}(\lambda) = \lambda^2 + \lambda - 2 = (\lambda - 1)(\lambda + 2)}$$

— **the anchor quadratic**: eigenvalues $\{1, -2\}$ = the distinct anchor roots, and the Nariai cubic factors as $(t-1)^2(t+2) = (t-1)\cdot\chi_{\rm clock}(t)$. The anchor thus appears a *third* time: as configuration roots, as the curvature base, and as the spectrum of the linearized clock. The horizon split is linear in the canonical angle with coefficient $1/\sqrt{N_{\rm fam}}$, and the entropy deviation grows at exactly $2H = |\mathbb{Z}_2|\times$Hubble. **Honest $(2/3)$-test [C]:** the classical clock eigenvalues are *integers*, not $(2/3)$-powers — the $(2/3)$-powers stay in the functional (curvature $(2/3)^{N_{\rm fam}}$). The conversion curvature→rate at one loop (quantum (anti-)evaporation) remains external QFT input: the bridge to the flavor rate $(2/3)^{2N_{\rm fam}}$ is *not* closed here. The seam variational principle now has exactly *one* missing object: the quantum clock. [verification/v104_nariai_clock.py]

The quantum-clock target, made quantitative ([E]/[C]). Instead of guessing the quantum clock, its job description can be pinned exactly. **(1) The classical clock already lives on the cusp ladder [E]:** across the physical Ginsparg–Perry modes ($l = 0$: $(\lambda{-}1)(\lambda{+}2)$; $l = 1$: $\lambda(\lambda{+}1)$) the decay exponents are

$$\boxed{\{0, -1, -2\} = -\,\mathrm{Spec}(V) = -N_{\rm fam} \times \text{cusp weights}}$$

— exactly the grading that carries the transfer operator's eigenspaces; the $l \geq 2$ modes are damped oscillations with $\mathrm{Re}\,\lambda = -\frac{1}{2} = -\delta$ (audit). **(2) The exact target [E]:** the transfer rates $\{0, \Delta, 6\ln 3\}$ are *not* weight-linear — rate(2)/rate(1) $= \log_{3/2} 3 = 1 + \log_{3/2} 2 = 2.7095$, with the exact identity $(1/3)^6 = \left((2/3)^6\right)^{\log_{3/2} 3}$: *the quantum clock must bend the weight-linear spectrum by* $\log_{9/4} 3 = 1.35476$ *per weight step.* **(3) Coupling landscape [E]:** the dimensionless clock coupling $\kappa = (c/24\pi)(\Lambda/\bar{M}_{\rm Pl}^2)$ with $c = 8$ is 7.6×10^{-122} at the cosmological Nariai (hopeless) and $\kappa = 1/(3\pi) = 0.106$ at seam curvature — the only regime where an $O(0.35)$ bend is reachable, and it is borderline non-perturbative: one-loop linear response cannot fix the bend; an $O(1)$ resummation or an exact seam construction is required. *Status [C]:* R1 is sharpened from "find a

clock" to "produce the bend $\log_{3/2} 3$ on the cusp ladder at coupling $O(1/3\pi)$". Nothing is fitted; the identification stays open. [verification/v107_quantum_clock_target.py]

The resummed clock: the bend has a closed form ([E]/[C]). The "bend" is not an exotic target — the frozen transfer spectrum *is* a resummed logarithm. **(1) Spectrum reading [E]:** the eigenvalues $\{1, (\frac{2}{3})^6, (\frac{1}{3})^6\}$ are exactly

$$\boxed{\lambda_n = \Big(1 - \frac{n}{N_{\rm fam}}\Big)^{p_2}, \qquad n = 0, 1, 2} \qquad (p_2 = 6 = |R^+(A_3)|),$$

so the clock has the closed form $\mathrm{rate}(n) = -p_2 \ln(1 - n/N_{\rm fam})$ — it reproduces $\{0, \Delta, 6\ln 3\}$, and the v107 bend $\log_{3/2} 3$ becomes a one-line consequence. **(2) The linearisation carries the sheet [E]:** $\mathrm{rate}(n) = |\mathbb{Z}_2|\, n + \frac{n^2}{3} + \frac{2n^3}{27} + \cdots$ — the one-loop term has slope exactly $|\mathbb{Z}_2| = 2$ relative to the *integer* classical Ginsparg–Perry clock $\{0, -1, -2\}$: the quantum clock is the sheet-doubled classical clock plus the geometric tail $\sum_k (n/N)^k/k$. **(3) The wall at $N_{\rm fam}$ [E]:** rate $\to \infty$ as $n \to N_{\rm fam}$ — the resummed logarithm has a pole at the family count, so the cusp ladder *must* end at $n = N_{\rm fam} - 1 = 2$: the three-level transfer spectrum is forced by the clock itself. *R1 re-sharpened [C]:* the semiclassical job is now "derive $-p_2 \ln(1 - n/N_{\rm fam})$ from the near-Nariai one-loop at $\kappa = 1/(3\pi)$" — a closed-form target, with the linear-response slope $|\mathbb{Z}_2|$ as its first checkpoint. All spectra frozen (v54/v69/v95/v104/v107); nothing fitted. [verification/v124_resummed_clock.py]

The clock and the wall share a spectrum ([E]/[C]). Both ends of the resummed clock connect to established exact objects. **(1) The weights are the parabolic weights [E]:** the arguments $n/N_{\rm fam} = \{0, \frac{1}{3}, \frac{2}{3}\}$ of the resummed logarithm are *exactly* $\mathrm{spec}\, A_0^\star$ — the spectrum of the exact $(U_{\rm wall})$ residue (research contracts):

$$\boxed{\lambda_n = (1 - \alpha_n)^{p_2}, \qquad \alpha_n \in \mathrm{spec}\, A_0^\star = \{0, \tfrac{1}{3}, \tfrac{2}{3}\}}$$

— the horizon clock and the flavor wall share their spectrum through the complement map $\alpha \mapsto 1 - \alpha$. **(2) Checkpoint 1 passes classically [E]:** the clock's linear coefficient $p_2/N_{\rm fam} = 2 = |\mathbb{Z}_2|$ *equals* the established classical entropy-deviation rate $2H = |\mathbb{Z}_2| \times$Hubble (v104) — the slope-2 linear response is already the classical entropy rate; the genuinely *quantum* content of R1 isolates to the geometric tail $\sum_{k\geq 2} \alpha^k/k$. **(3) The determinant is the entropy curvature [E]:** $\det(\mathbf{1} - A_0^\star) = \frac{2}{9} = |\mathbb{Z}_2|/N_{\rm fam}^2$ — exactly the gravity entropy curvature at the anchor (v102), a fourth independent appearance; $\mathrm{tr}(\mathbf{1} - A_0^\star) = 2 = |\mathbb{Z}_2|$, $\det A_0^\star = 0$. *R1 re-targeted [C]:* derive "rate $= -p_2 \ln(1 - \alpha)$" with α the Mehta–Seshadri parabolic weight — first order classical, tail quantum: **one spectrum, two gates, one object** $A_0^\star$. [verification/v126_clock_wall_bridge.py]

The clock is a log-determinant: the tail is the ring (RPA) resummation ([E]/[C]). The "$O(1)$ resummation" demanded by v107 is now *typed*. For a rank-one insertion P ($\mathrm{tr}\, P^k = 1$), $-\ln\det(\mathbf{1} - \alpha P) = -\ln(1 - \alpha) = \sum_k \alpha^k/k$, so the resummed clock is exactly

$$\boxed{\mathrm{rate} = -p_2\ \mathrm{Tr}\ln(\mathbf{1} - \alpha P)}$$

— $p_2 = 6$ *identical* one-loop ring (RPA) towers, **one per hexagon site** (= positive root of A_3), with coupling $\alpha =$ the parabolic weight. *Ring ladder [E]:* the k=1 ring is the *classical* term ($6\alpha = |\mathbb{Z}_2| n$, the v104/v126 entropy rate); the k=2 ring is the first quantum correction ($\frac{1}{3}$ at weight 1); the partial sums converge to $\{\Delta, 6\ln 3\}$ monotonically from below. *Coupling cross-link [E] (audit):* the established seam coupling is $\kappa = \frac{c}{24\pi} = \frac{8}{24\pi} = \frac{1}{3\pi} = \alpha_1/\pi$ — the *first parabolic weight over* π. *The wall [E]:* the ring series diverges at $\alpha \to 1$ ($n \to N_{\rm fam}$) — the v124 wall is the standard RPA instability. *R1 final form [C]:* show that the near-Nariai one-loop effective action of the seam mode, with insertion = parabolic weight and multiplicity p_2, is this ring sum — "**the seam clock is the RPA determinant of one mode on six sites**". The resummation type is identified; the derivation itself remains open.

Where the six sites live ([verification/v128_graded_hull.py]): the Ginsparg–Perry $l = 1$ sector — the

zero modes $\lambda \in \{0, -1\}$ of the classical clock (v104/v107) — has $2l+1 = 3$ modes, and sheet doubling gives $2 \times 3 = 6 = p_2$ [E]: the ring multiplicity equals the sheet-doubled zero-mode count. [C] reading (recorded): zero modes are exactly the modes that force nonperturbative resummation in de Sitter one-loop physics — the natural home of the six RPA towers. [verification/v127_ring_resummation.py]

The clock is an entropy power law ([E]/[C]). What the resummation argument $1 - n/N_{\text{fam}}$ *is* physically: the three established Nariai entropy levels (v101) are $S/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$ (pure dS; two-horizon total; single horizon — deficit $S_{dS}/3$ per step) — exactly the complementary cusp weights. Hence [E]:

$$\boxed{\lambda_n = \left(\frac{S_n}{S_{dS}}\right)^{p_2}, \qquad \text{rate}(n) = p_2 \ln \frac{S_{dS}}{S_n}}$$

— the transfer eigenvalues are the *entropy fractions to the hexagon power*; the rates are hexagon-weighted log-entropy deficits. *The coupling is the entropy deficit [E] (triple identity):* $\alpha = n/N_{\text{fam}} =$ the Mehta–Seshadri parabolic weight (v115/v126) $= 1 - S_n/S_{dS} = \Delta S_n/S_{dS}$ — the RPA insertion strength is the fractional entropy deficit; the coupling is not a free parameter in *any* of its three readings. *Orientation [E]:* weight $0 \leftrightarrow$ pure dS (maximal entropy, stationary attractor), weight $2 \leftrightarrow$ single horizon (fastest decay) — the established repeller→attractor flow. *R1, thermodynamic form [C]:* derive $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ from the near-Nariai one-loop — a Gibbons–Hawking-type entropy power law with exponent = the hexagon size (audit: $p_2 = 6 = 2h$ reads as the two-point form of a weight $h = 3 = N_{\text{fam}}$ operator; recorded, not claimed).

The Born square ([verification/v130_born_square.py]): the weight audit is now *derived* from mode counting [E]. In the standard collective-coordinate treatment each zero mode contributes a moduli factor $\propto S^{1/2}$, so with $n_{\text{zero}} = 6$ the transition *amplitude* scales as $A \sim (S/S_{dS})^3$ and the *probability* is its Born square $\Gamma = |A|^2 = (S/S_{dS})^6$:

$$\boxed{h = \tfrac{n_{\text{zero}}}{2} = 3 = N_{\text{fam}}, \qquad p_2 = 2h \text{ (the Born rule)}}$$

— and the 3 is the dimension of the l=1 eigenspace = the number of *proper conformal Killing vectors of* S^2 (the classic near-Nariai zero modes, Ginsparg–Perry/Bousso–Hawking class), while the $\times 2$ is the *horizon pair* = the SdS mass-line deck (v101). The same 3 counts the generations and the sphere's conformal directions. *Residue [C]:* justify the per-zero-mode $S^{1/2}$ normalisation on the Nariai instanton for exactly these six modes — a textbook-class computation; no exotic object remains in R1.

The measure is the area ([verification/v131_measure_is_area.py]): the geometric core of that normalisation is exact [E]: on $S^2(r)$ the L^2 norm of *each* of the three normalised l=1 harmonics is

$$\boxed{\|Y_{1m}\|^2 = r^2 = \frac{A}{4\pi}}$$

(all three integrated exactly) — the zero-mode norm *is* the horizon area, hence the entropy. The collective-coordinate Jacobian per mode is the mode norm $\sim S^{1/2}$ (standard change of variables), so: norm = area → Jacobian $= S^{1/2}$ → six modes → amplitude S^3 → Born → S^6 — every exponent factor carries a derivation-level origin. The 4π is the same Gauss–Bonnet primitive that builds $c_3 = \frac{1}{2}(4\pi)^{-1}$, and it *cancels* in all ratios — exactly why the clock sees only entropy fractions. *The last step [C]:* show $\det'_n / \det'_{dS} = 1 + O(\text{deficit})$ (non-zero-mode determinant cancellation; same local near-Nariai geometry). [verification/v129_entropy_power_law.py]

The cancellation, exact within the family ([verification/v144_detratio_family.py]): that step is now *derived* in the within-family sense [E]. The traceless SdS cubic forces

$$\boxed{r_b\, r_c = 1 - \frac{\Delta^2}{3} \quad (\Delta = r_b - r_c \text{ the horizon split; exact, no higher corrections})}$$

(e_2-rigidity), and with the v132 anomaly $\zeta(0)|_{\det'} = -\frac{2}{3}$ per sphere the two-sphere non-zero-mode determinant ratio against Nariai is the closed form $\det'_n / \det'_N = (1 - \Delta^2/3)^{4/3}$: **no first-order term in the split** — the hoped-for cancellation is a consequence of e_2-rigidity plus

the established anomaly, not an assumption. In deficit form, $\varepsilon = 1 - c = \frac{3}{8}\Delta^2 + O(\Delta^4)$ gives $\det'_n / \det'_N = 1 - \frac{32}{27}\varepsilon + O(\varepsilon^2)$ with $\frac{32}{27} = |\mu_4|(\frac{|\mathbb{Z}_2|}{N_{\text{fam}}})^3$ (the v103 canonical curvature times the glue order; audit-typed). *Honest scope [C]:* the finite-weight comparison ($S_n/S_{dS} \in \{1, \frac{2}{3}, \frac{1}{3}\}$) still runs through the v132/v133 budget — a naive global-rescaling reading would shift the Born exponent $6 \to \frac{14}{3}$ (not an atom), so the absorption must proceed via the Hubble-unit conversion; the frozen transfer spectrum is untouched, and the remaining [C] now has its obstruction stated sharply.

The clock is a Gaussian zero-mode integral (`[verification/v147_clock_gaussian.py]`): the v127 typing job is discharged at the model level, and the bend is $\det'$-clean [E]. For p_2 Gaussian zero modes whose per-mode variance scales by $(1-\alpha)$ — *forced* by norm = area (v131) — the Born-squared partition ratio is exactly $(1-\alpha)^{p_2}$, i.e. rate $= -p_2 \ln(1-\alpha)$, and the ln series *is* the ring sum $p_2 \sum_k \alpha^k/k$ term by term. Sharper still: the two quantum weights $n = 1, 2$ are bookkeeping levels of *one* Euclidean geometry (Nariai $S^2 \times S^2$, v101), so $\det'_1 = \det'_2$ *identically* — **the bend $\log_{3/2} 3$ (the v107 target) carries no determinant correction at all**. The single across-topology step is the $\alpha = 0$ reference (round dS vs Nariai), exactly where the v132/v133 budgets and the v144 obstruction live; its classical side is already matched (v126: slope $2 = |\mathbb{Z}_2|$). *What stays [C]:* the measure identification ("the gravitational near-Nariai zero-mode measure *is* this Gaussian model") and the reference normalisation — structure, spectrum, bend and ring typing are exact.

The det-ratio anomaly is the Koide constant ([E]/[C]). The det-ratio question acquires its first *exact* invariant. The Ginsparg–Perry sphere operator on the unit S^2 is $L = -\Delta - 2$ (the $-2 = -|\mathbb{Z}_2|$ is the v104 modulus mass), with eigenvalues $(l+2)(l-1)$ and kernel the three $l{=}1$ zero modes. Its heat coefficient is made of atoms [E]: $a_1 = \frac{R}{6} - E = \frac{1}{3} + 2 = \frac{1}{N_{\text{fam}}} + |\mathbb{Z}_2| = \frac{7}{3}$, and the scaling anomaly of the non-zero-mode determinant is

$$\boxed{\zeta(0)\big|_{\det'} = a_1 - \dim\ker = \tfrac{7}{3} - 3 = -\tfrac{2}{3} = -\frac{|\mathbb{Z}_2|}{N_{\text{fam}}}}$$

— confirmed independently by numerical continuation of the heat trace ($\sim 10^{-7}$): **the det-ratio anomaly is the Koide constant**, the eighth appearance of 2/3 and the first as a *spectral* anomaly. *Consequence [E]:* $\det'(cL) = c^{\zeta(0)} \det'(L)$ — the determinant ratio between configurations is *not* automatically 1; the S^2-sector share of the scaling budget is exactly $-\frac{2}{3}$ per Hubble-unit conversion, while the dimensionless tower itself is configuration-independent (pure-number GP spectrum). *The budget statement [C] (the remaining step, made precise):* show the *full* $\zeta(0)$ budget of the 4d Nariai fluctuation problem (dS_2 + gauge/ghost + this S^2 sector) vanishes or is absorbed into the established power law — the S^2 share is pinned exactly; nothing is claimed about the total. `[verification/v132_detratio_anomaly.py]`

The zeta budget: the reduced route carries the anchor ratios ([E]/[C]). Euclidean Nariai is $S^2 \times S^2$ — the temporal sector continues to the *same* unit sphere, so both candidate determinant routes are exactly computable. *(i) The reduced route [E]:* each 2d sector carries the v132 operator with $\zeta(0)|_{\det'} = -\frac{2}{3}$; the two sectors give

$$\boxed{\text{budget}_{\text{red}} = -\tfrac{2}{3} - \tfrac{2}{3} = -\tfrac{4}{3} = -\frac{e_1}{p_0} \text{ (minus the seed gain)}}$$

— the per-sector share and the total are *two of the three v119 anchor-ratio triad values*; the six zero modes split $3 + 3$ across the two spheres: the sheet doubling *is* the two-sphere structure of Euclidean Nariai (= the horizon pair, v101). *(ii) The naive 4d route [E]:* with the standard S^2 heat expansion $1/t + \frac{1}{3} + t/15$ and the symmetric split, $a_2(4d) = (\frac{4}{3})^2 + 2 \cdot \frac{9}{10} = \frac{161}{45}$, six zero modes, $\zeta(0)|_{\det',4d} = \frac{161}{45} - 6 = -\frac{109}{45}$ — *no* anchor-atom reading. *(iii) Route discrimination [C]:* the budget computation structurally favours the **reduced seam reading** (pure anchor-ratio anomalies), consistent with the theory's 2d-seam architecture throughout — route-selection evidence, not proof. *Remaining [C]:* the graviton/ghost shares on $S^2 \times S^2$ (Volkov–Wipf class, external standard) — a finite list of known heat coefficients. `[verification/v133_zeta_budget.py]`

The Volkov–Wipf firewall: the external edge is two copies of the reduced budget ([E]/[C]). The "finite list" above is now *filled in externally.* For the full one-loop graviton path integral on $S^2 \times S^2$ the log-coefficient is (arXiv:2506.02142, Table 4.1, agreeing with Volkov–Wipf, Nucl. Phys. B582 (2000) 313):

$$\alpha_{\rm bulk} = \frac{22}{45}, \qquad \alpha_{\rm edge} = -\frac{8}{3}, \qquad \alpha_{\rm PI} = -\frac{98}{45}$$

(no anchor atom; the difference to the reduced budget is $-\frac{38}{45}$ with $38 = 2 \cdot 19$). The structural finding: the external *edge* factor consists of **two identical copies** (one per horizon), and

$$\boxed{\alpha_{\rm edge} = -\frac{8}{3} = 2 \times \left(-\frac{4}{3}\right)}$$

— the per-copy magnitude $\frac{4}{3} = e_1/p_0$ is *exactly* the reduced seam budget above (the seed gain; $-\frac{2}{3}$ per sector). The TFPT reduced route lands on the **edge sector** of the external bulk/edge split; the bulk ($\frac{22}{45}$) is the ideal graviton gas, which the seam clock does not contain; two copies = the horizon pair = the sheet doubling (v101; the 3+3 zero-mode split). *Typing decision [C] (recorded):* R1 = the reduced zero-mode/RPA seam clock = an edge-sector object; the full 4d Volkov–Wipf determinant is the *gravity firewall* (its $-\frac{98}{45}$ is absorbed in the $\mu_0/(\Lambda G)^3$ normalisation), not a TFPT atom. Per-copy sign/orientation (the edge factor enters inverted) and the Lorentzian edge-mode reading stay [C]. [verification/v138_vw_firewall.py]

Black-hole mechanics in seam units ($c_3 = 1/(8\pi)$): every coefficient is a compiler atom

quantity	standard form	seam form	compiler atom
Einstein equations	$G_{\mu\nu} = 8\pi T_{\mu\nu}$	$G_{\mu\nu} = T_{\mu\nu}/c_3$	$c_3 = 1/(8\pi)$ (P1)
first law	$dM = \frac{\kappa}{8\pi} dA$	$dM = c_3\,\kappa\, dA$	seam constant
Smarr (Schw.)	$M = \frac{\kappa A}{4\pi}$	$M = 2c_3\,\kappa A$	
temperature	$T_H = \frac{\kappa}{2\pi}$	$T_H = 4c_3\,\kappa$	$1/(2\pi) = 4c_3$
entropy	$S = \frac{A}{4}$	$S = 2\pi c_3 A$	$1/4 = 1/\lvert\mu_4\rvert$
Bekenstein bound	$S \le 2\pi ER$	$S \le ER/(4c_3)$	
Hawking power	$P = \frac{1}{15360\,\pi M^2}$	$P = \frac{c_3}{1920\,M^2}$	$1920 = \lvert W(D_5)\rvert$
lifetime	$\tau = 5120\,\pi M^3$	$\tau = 128\,g_{\rm cat} M^3/c_3$	$128 = 2^7$, $7 =$ scalaron
Kerr extremal	$A_{\rm ext} = 8\pi M^2$	$A_{\rm ext} = M^2/c_3$	$A_{\rm ext}/A_{\rm Schw} = 1/\lvert\mathbb{Z}_2\rvert$
area quantum (Hod)	$\Delta A = 4\ln 3$	$\Delta A = \ln(N_{\rm fam}^4)$	$81 =$ disc of the cover
Nariai bound	$S_N = \frac{2\pi}{\Lambda}$	$S_N = \frac{2}{3} S_{dS}$	$2/3 = \lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ (Koide)

Figure 6: Black-hole mechanics in seam units: every classical coefficient is a compiler atom. [verification/v101_horizon_anchor.py]

6 Turnaround radius, gravitational waves, light

Bound structures. In a Λ-universe the maximal bound radius is $r_{\rm ta}(M) = \left(GM[\,2\pi e^{\alpha_\star^{-1}}/\bar{M}_{\rm Pl}]^2\right)^{1/3}$ — halo, cluster, supercluster and void scales follow from the Λ closure with no new parameter. [C]

Gravitational waves. The low-curvature branch is $R + R^2$; the tensor graviton runs on the Lorentz cone and the only extra mode (the scalaron, $M_{\rm scal} \approx 3.06 \times 10^{13}$ GeV) is frozen for astrophysical frequencies. Hence

$$\boxed{v_{\rm GW} = c} \qquad \text{(no measurable dispersion)} \qquad \text{[C]},$$

a clean falsifier: a genuine $v_{\rm GW} \neq c$ would break this gravity closure.

Light. TFPT does not predict the SI number c (definitional); it predicts the *universality of the causal cone* — RP → OS reconstruction → one Lorentzian cone shared by EM, gravity and massless modes ($v_\gamma = v_g = c$, no native photon dispersion). What it *does* compute is not a speed shift but a polarisation rotation, cosmic birefringence

$$\boxed{\beta_{\rm rad} = \frac{u}{4\pi} \approx 0.2424^\circ} \qquad \text{[E]},$$

from the same seed $u = \phi_0^{\rm ret}$ that fixes the Cabibbo angle, Ω_b and θ_{13}.

7 The local sibling: an EHT achromatic polarization intercept

The cosmic birefringence $\beta_{\rm rad}$ is the *asymptotic* (Thomson-limit) projection of the determinant-line response; the *local* projection onto the magnetised collar of a horizon-scale black hole is a structured, achromatic residual rotation of the linear-polarization angle [verification/v203_eht_achromatic.py]

$$\beta_{\rm BH}(r) = \frac{Q_e^{\rm eff}(x)\, Q_m^{\rm eff}(x)}{256\pi^4\, r^2} = 16\, c_3^4\, \frac{Q_e^{\rm eff} Q_m^{\rm eff}}{r^2} = \frac{\delta_{\rm top}}{3}\, \frac{Q_e^{\rm eff} Q_m^{\rm eff}}{r^2},$$

where the coupling $16c_3^4 = 1/(256\pi^4)$ is fixed *exactly* ([E]) — the *same* top-form coefficient $\delta_{\rm top} = 48c_3^4 = 3/(256\pi^4)$ that controls the α-kernel precision-zone correction. There is *no free coupling*: EHT-class polarimetry probes the same admissibility data as the α row, only projected onto the local collar. What TFPT fixes is the structured $1/r^2$ shape, the achromaticity, and the sign-flip under effective $E{\cdot}B$ reversal, *not* the amplitude $Q_e^{\rm eff} Q_m^{\rm eff}$ (an MHD/GR weight); the channel is therefore typed [C] (shape/sign), not a parameter-free pixel prediction.

Dated kill test [X]. Build the residual intercept $\chi_0^{\rm res} = \chi_0^{\rm obs} - \chi_0^{\rm GRMHD}$ after honest GRMHD/synchrotron subtraction; it must pass *three independent nulls* — frequency (achromatic, no λ^2 tail), spatial ($1/r^2$ profile), and sign-flip (reverses under $E{\cdot}B$ reversal). A residual statistically consistent with zero across the horizon-scale image, or failing any null, falsifies *this channel* (the compiler core is untouched). *Scope:* the old UFE closed RN-type metric (D/r^4, regular cores, shadow shift) is *not* adopted — it is superseded by the Nariai/seam=horizon readouts above (v190); only the polarization signature survives.

8 Search targets (not claims)

Audit-level search ansätze [O]

- **Black-hole echoes / horizonless compactness:** any near-horizon echo amplitude ratio would be a natural transfer-rate target, $\mathcal{A}_{n+1}/\mathcal{A}_n \lesssim (2/3)^6 \approx 0.0878$. The *same* window $Q_{\rm geom} \in [\frac{3}{8}, \frac{1}{2}]$ that the SdS roots trace surfaces independently as the maximum horizonless gravastar compactness $\mathcal{C} = \frac{3}{8}$ (de Sitter endpoint $\frac{1}{2}$) of Jampolski–Rezzolla (2026); since $\frac{1}{3} < \frac{3}{8} < \frac{4}{9} < \frac{1}{2}$ such an object is a horizonless light-trapping ECO. `experiments/gravastar-compactness` turns this into an echo template (round-trip delay ~ 0.7 ms at $62\,M_\odot$, amplitude $\leq (2/3)^6$) — a [C] structural echo, no $\mathcal{C} \leftrightarrow Q_{\rm geom}$ map proven. A search ansatz, no prediction.
- **Quantum recovery:** the Page-curve recovery kernel $I \sim (2/3)^{6n}$ is a falsifiable shape if a boundary-recovery rate is ever measured.
- **BH HFQPO ladder tooth (×1.5):** the four published black-hole HFQPO twin pairs are consis-

tent with exactly 3:2, but that cluster is *cheap* (an anchored detection-selection null reproduces it in 18.5% of trials, and XTE J1859+226 breaks universality at $+9.2\sigma$), and mapping the TFPT relaxation-ladder step 3/2 (from $\text{rate}(n) = -6\ln(1 - n/N_{\text{fam}})$, $N_{\text{fam}} = 3$) onto a two-oscillator frequency ratio is *non-canonical* — GR parametric resonance is the standing favorite. The one discriminating target is a *third tooth* at $\nu_3 = (3/2)\,\nu_u$ (661.5/414/252/363 Hz for the four sources) with integer harmonics forbidden on the geometric ladder. No published search ever targeted these frequencies; the preregistered archival RXTE PCA scan (executed 2026-07 on 77 ObsIDs across the four sources; sanity gate 11/12 published pair lines reproduced, injection-calibrated in every source) finds *neither* the tooth *nor* the integer control line $4\nu_0$ anywhere ($\sim 0\sigma$, trials-corrected), with 3σ rms upper limits of 0.53–3.06% — in every source at or below the strength of the detected upper pair line. A well-powered null: the ladder reading gains no support (GR resonance stays favored) but is not killed, since the harmonic branch did not fire either (the published integer lines 92=184/2, 34/68 Hz stand). `experiments/hfqpo-ladder` carries the preregistered census, selection-null Monte Carlo and the executed archival scan. A search ansatz, no prediction; even a future tooth hit would be [C] until the mapping is derived.

- **FRB repeaters:** fast-radio-burst repeaters are a *preregistered search interface* for the same boundary-recovery kernel (the frozen ratios $\{1, (2/3)^6, (1/3)^6\}$ and $\{1, (2/3)^3, (1/3)^3\}$), *not* a horizon claim and *not* part of the verification ledger — the work lives in `experiments/frb-tfpt-signatures` with a frozen hypothesis file and surrogate-calibrated null tests; current public data neither confirm nor refute the kernel (a clean null on echo ratios, a single non-replicated window candidate). Hunting ground, not foundation. A joint five-year *power forecast* across the comb channels (PSR/FRB/GW/UHECR/CMB and siblings) now lives in `experiments/p9-forecast` (preregistered before any Monte-Carlo run, anchored to the published sibling calibrations; verdict P9-FORECAST-GO on two channels plus the hierarchical combination, with its three load-bearing conditionals named) — a forecast, *no* data analysed, *no* evidence row, firewalled exactly like the siblings.
- **Cosmological coupling ($w = -1$):** if the de Sitter–vacuum seam interior (the regular-core reading above) makes a seam black hole carry the Λ vacuum, $w_{\text{in}} = -1$, then under Croker–Weiner cosmological coupling $k = -3w_{\text{in}}$ the mass-growth index is $k = 3$ exactly, i.e. a population $w = -1$. `experiments/ccbh-dark-energy` confronts this with Farrah+2023 ($k = 3.11 \pm 0.79$, -0.14σ) as a *contested* downstream bridge and an *alternative reading* of the same $w = -1$ the DESI w-watchdog tests — not two independent hits.
- **Cosmic spin parity (frontier):** the μ_4 clock and the $PSL(2,\mathbb{C}) \cong SO^+(3,1)$ boundary orientation allow at most a tiny global remnant, so TFPT predicts *approximate parity*, not a large handedness excess. `experiments/cosmic-handedness` watches the galaxy-spin asymmetry (Shamir JADES $\sim 3.3\sigma$ monopole) — most likely a Milky-Way-rotation *dipole* systematic and isotropic in Galaxy Zoo — a frontier watchdog, explicitly *not* promoted.

Net

All horizon formulas share the one seam normalisation $1/(2\pi) = 4c_3$. Black-hole, de Sitter and Unruh temperatures, the Page time, scrambling, the Nariai bound, turnaround radii, $v_{\text{GW}} = c$ and birefringence are *readouts of one seam constant* — standard physics rewritten in TFPT units, with two genuine compiler fingerprints (1920 = $|W(D_5)|$ in Hawking power, $|\mu_4| = 4$ in scrambling). No new gravitational physics is claimed; the value of the reframe is that horizons stop being separate modules.

Appendix: computational verification

The seam-unit identities of this note (the thermal grammar $\frac{1}{2\pi}$=$4c_3$, the power denominator 1920=$|W(D_5)|$, the de Sitter entropy form, the Page fraction, the scrambling $|\mu_4|$=4, and β_{rad}=$\varphi_0^{\text{ret}}/4\pi$=0.2424°) are re-derived and machine-checked in `verification/v8_horizon.py`. Run `python verification/v8_horizon.py` (needs `mpmath`); it imports the same c_3=$1/(8\pi)$ primitive as the rest of the suite. The search-target ansätze remain [O] and are deliberately not part of the pass/fail checks.

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

Modular theory and algebraic quantum field theory.

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- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

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- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, `arXiv:2107.13834`, `doi:10.1007/s00220-022-04521-8`. (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur → Virasoro convergence, $c = \frac{1}{2}$, and the smeared WZW-current Theorem 5.1. The internal TFPT code-token "MMST" and the legacy claim-ids `SEAM.MMST.*` / filenames `v*_mmst_*` denote *this* Osborne–Stottmeister theorem.)
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- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, `arXiv:2407.18222` (2024).
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